Technology Stack

Date	27 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Used to build a responsive and user-friendly interface where farmers can easily enter soil and weather details and view crop recommendations.

2	Backend / Server-Side	Python, Flask	Flask handles user requests, processes soil and environmental data, and connects the machine learning model with the web application.
3	Database / Data Storage	CSV Dataset, Pandas	Stores crop and soil information used for prediction. Pandas efficiently reads and processes agricultural datasets for analysis.
4	Cloud / Hosting / Deployment	Vercel (Frontend), Flask Backend	The frontend is deployed on Vercel for easy access and fast delivery, while the Flask backend serves prediction requests generated by the machine learning model.

5	Version Control & CI/CD	GitHub	Tracks project changes, enables collaboration between team members, and maintains source code versions.
6	Third-Party APIs / Other Tools	Scikit-learn, NumPy, Matplotlib	Scikit-learn is used for crop prediction, NumPy performs numerical calculations, and Matplotlib helps visualize agricultural data and model results.